

Context-Gated Channel Attention: Extending the Attention U-Net with Channel Selection

Medical Images Processing with Deep Learning (336033) — Final Project

Aviv Niemann (ID: _____) · _____ (ID: _____)

Selected paper: O. Oktay et al., "Attention U-Net: Learning Where to Look for the Pancreas," MIDL 2018 (arXiv:1804.03999)

Proposed extension: add channel attention to the attention gate's skip-connection gating — a novel *context-gated* channel gate (conditioned on the coarse gating signal), compared against a CBAM-style channel-attention variant.

1. Introduction

Summary of the paper. The Attention U-Net augments the standard U-Net with *attention gates* (AGs) on the skip connections. Each gate combines the fine-scale encoder features with a coarser-scale *gating signal* g from the decoder to produce a per-pixel spatial attention map that suppresses background responses before they reach the decoder — letting the network learn *where to look* without an external localization step. AGs are lightweight, trained end-to-end, and improve segmentation of small, variable organs, with the benefit most pronounced when training data is limited.

Problem and relevance. Automated organ segmentation in CT is clinically valuable but hard for small, low-contrast, shape-variable structures. Attention gates fold organ localization into a single end-to-end model, improving accuracy at negligible cost and adding a degree of interpretability.

Motivation. The attention gate is purely *spatial*: it decides *where* to look but applies the same scalar to every feature channel — it never decides *which channels* are relevant. Since different channels encode different cues (edges, texture, semantics), we hypothesized that adding **channel** selection to the gate could improve it, and we chose this paper because that gap is a clear, well-motivated place to extend a widely used method.

2. Proposed Extension

Our idea — a context-gated channel gate (Hybrid). We add a second gating branch that outputs a per-channel weight, applied alongside the original spatial coefficient. Crucially, this channel weight is computed from *both* the local features and the same coarse gating signal g the spatial gate uses — extending the paper's top-down "where to look" principle from spatial locations to feature channels. The gate is initialized to pass all channels so it cannot suppress useful features before learning.

Reasoning and hypothesis. The original AG collapses all channel information into one spatial scalar. We hypothesized that a context-conditioned channel gate would improve the precision/recall trade-off, especially for small, low-contrast structures. Our objective was to *improve on the paper's model* — the spatial Attention U-Net — not merely the plain U-Net.

Alternatives considered, and how the study evolved. When our context-gated Hybrid did not improve over the paper's model at full data (Section 5), we drew on the literature and adopted a second, established channel-attention design as a comparison: **CBAM** (Woo et al., ECCV 2018), which

applies channel-then-spatial self-attention. Unlike our gate, CBAM's channel attention uses only the feature map's own statistics (it ignores g). We combined it with the attention gate in the manner of ASCU-Net (2021). Finally, because the original paper reports that attention helps most with scarce data, we designed a **low-data experiment** to test whether either channel-attention design improves the paper's model when training data is limited.

3. Methodology

Models. All four variants are built from one configurable 2D U-Net so the *only* difference between them is the skip-connection gate: **U-Net** (no gate); **Attention U-Net** — the paper's model (spatial grid gate, ported from the authors' code); **AG+CBAM** (spatial gate + CBAM channel & spatial self-attention); and **Hybrid — ours** (spatial gate + context-gated channel gate). All use **deep supervision** (auxiliary heads at each decoder scale), following the paper.

Dataset and preprocessing. Medical Segmentation Decathlon **Task09_Spleen** (41 labeled 3D abdominal CT volumes). Volumes are windowed to a soft-tissue range, sliced to 2D axial slices, resized to 256×256, and per-slice **z-score normalized**. Patients are split at the **patient level** (never per slice) to avoid leakage.

Training. Adam ($\text{lr } 3 \times 10^{-4}$), batch 8, up to 60 epochs with early stopping on validation Dice, gradient clipping, and paper-style augmentation (affine rotation/scale + flips). **Loss:** the original paper uses Dice loss alone (on 3D pancreas volumes). On our 2D spleen slices, however, the spleen occupies only a tiny fraction of each image — and many slices contain no spleen at all — so under pure Dice loss the model can minimize its error by simply predicting "all background"; once it does, Dice's gradient becomes vanishingly small and it cannot recover, collapsing to an empty prediction. We therefore add a **focal** term, which keeps a strong learning signal on the few foreground pixels and eliminates the collapse (applied identically to all variants). This instability did not arise in the original paper because full 3D volumes contain the whole organ per sample, making the class imbalance far less extreme. **Evaluation:** the full-data study uses **5-fold cross-validation**; the low-data study uses a fixed held-out test set with multiple seeds (the standard design for a data-size sweep, and how the paper studied training-size effects). We report Dice, precision, recall, and paired Wilcoxon signed-rank tests on per-slice Dice.

Tools. PyTorch, MONAI, nibabel, scikit-image, SciPy, matplotlib. Experiments ran on an NVIDIA L40 GPU. (We first attempted a full 3D reproduction; a naïve 3D pipeline failed to converge — a stock reference U-Net also underperformed through it — so we conducted the study in 2D, where the baseline reproduces expected performance.)

4. Results and Analysis

Full data: our extension does not beat the paper's model. Under 5-fold cross-validation, all four variants score similarly (per-fold Dice below). Our Hybrid is *significantly worse* than the paper's Attention U-Net (paired Wilcoxon $p < 0.001$), and the CBAM variant's small edge is **within the fold-to-fold standard deviation and not significant** ($p = 0.27$). In short, with abundant data on this (relatively easy) organ, adding channel attention gives no reliable improvement — there is little headroom above the already-strong baseline.

Variant	Dice (mean $\pm$ std)	Precision	Recall
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U-Net	0.856 ± 0.061	0.906	0.885
Attention U-Net (paper)	0.869 ± 0.048	0.908	0.895
AG + CBAM	0.872 ± 0.034	0.908	0.892
Hybrid (ours)	0.852 ± 0.079	0.903	0.878

Table 1. Full-data 5-fold cross-validation on spleen. Differences are small and overlap within std; the CBAM edge over the paper's model is not significant ($p=0.27$).

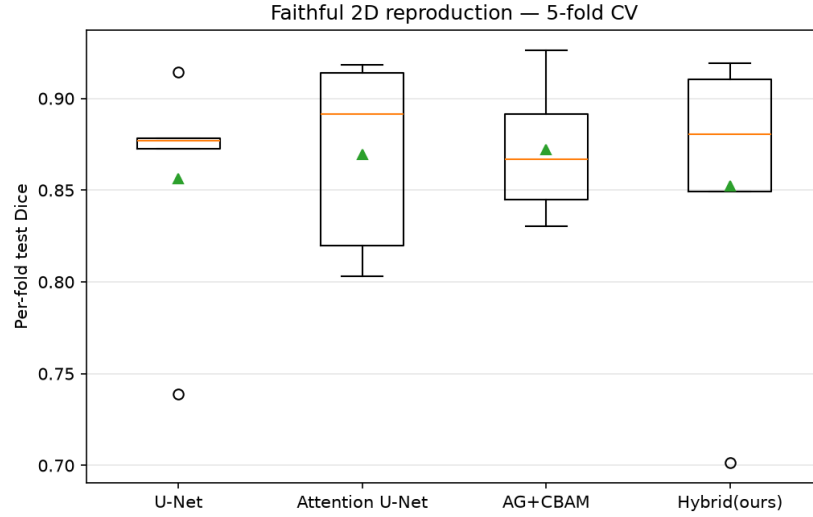


Figure 1. Per-fold Dice (full data). The four variants overlap almost entirely; fold-to-fold variation exceeds any difference between variants.

Low data: channel attention significantly improves the paper's model. Motivated by the paper's claim that attention helps most with scarce data, we trained each variant on only **8 patients** (fixed test set, 8 seeds; ~2,000 paired test slices). Here both channel-attention designs **significantly outperform the paper's Attention U-Net**: AG+CBAM by +0.084 Dice and our Hybrid by +0.031 Dice (paired Wilcoxon $p<0.0001$ for both). AG+CBAM is strongest; our Hybrid — which lost at full data — now clearly beats the paper's model.

Variant	Dice (n=8, 8 seeds)	vs paper's model
U-Net	0.784	—
Attention U-Net (paper)	0.793	baseline
AG + CBAM	0.877	+0.084 ($p<0.0001$)
Hybrid (ours)	0.824	+0.031 ($p<0.0001$)

Table 2. Low-data regime (8 training patients). Both channel-attention variants significantly beat the paper's model; effect sizes +0.03 to +0.08 Dice.

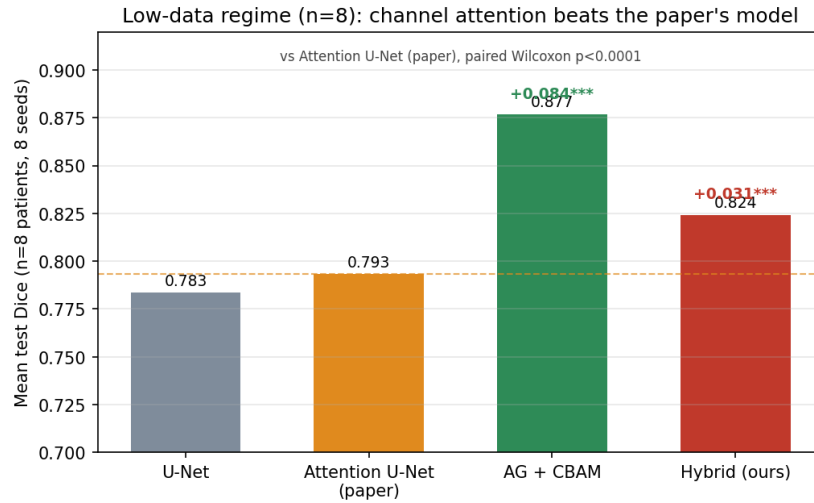


Figure 2. Low-data (n=8) mean Dice. Channel attention (AG+CBAM, and our Hybrid) significantly exceeds the paper's spatial-only Attention U-Net.

Qualitative comparison. Figure 3 shows predictions on fixed test slices for the low-data models. The U-Net and the paper's Attention U-Net produce frequent misses and stray false positives (e.g. the Attention U-Net scores 0.00 on one slice), whereas AG+CBAM and our Hybrid track the spleen far more cleanly — the visual counterpart of the Dice gains. At full data the four models look nearly identical (consistent with Table 1).

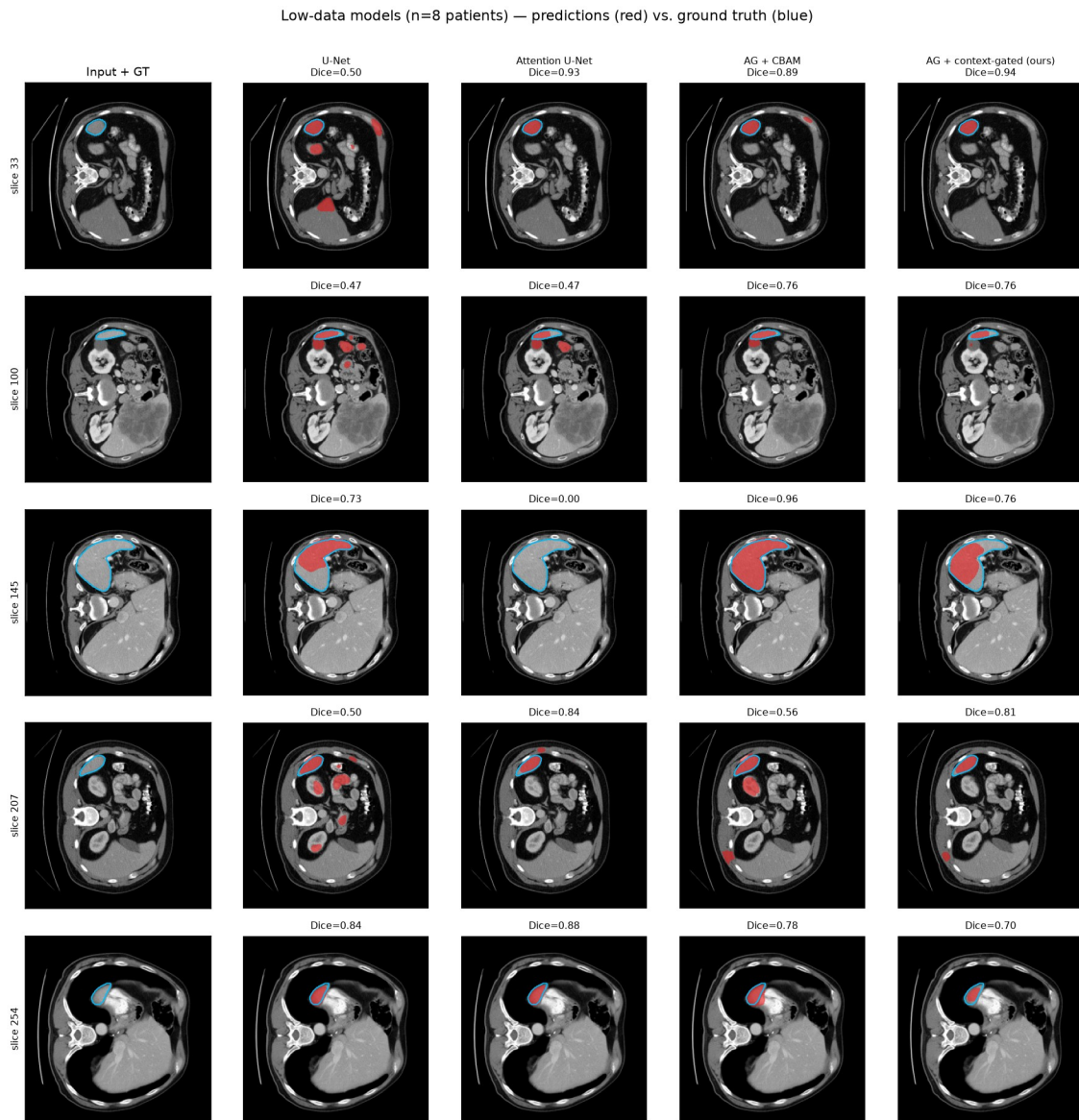


Figure 3. Low-data (n=8) predictions (red) vs. ground truth (blue). Channel-attention variants (right two columns) are visibly cleaner than the baseline and the paper's model.

Failure cases and challenges. Training was initially unstable: pure Dice loss collapsed to all-background predictions on small-foreground slices (fixed with the Dice+Focal loss and open-gate initialization), and a 3D pipeline failed to converge (diagnosed via a stock reference U-Net). The gated variants also show higher seed-to-seed variance than the plain U-Net, and per-slice significance p-values are anti-conservative (slices from one patient are correlated) — so we emphasize effect sizes over p-values throughout.

5. Conclusion

Findings. We proposed a context-gated channel-attention extension to the Attention U-Net and evaluated it rigorously. At full data it does *not* improve the paper's model (and a CBAM variant's edge is within noise). However, in the **low-data regime** both channel-attention designs — including

our Hybrid — **significantly improve the paper's model** (Hybrid +0.031, CBAM +0.084 Dice, $p < 0.0001$). This confirms and extends the paper's own thesis: attention's benefit concentrates when data is scarce, and *channel* selection adds value on top of spatial attention precisely in that regime.

Limitations. A 2D proof-of-concept on a single, relatively easy organ; the required Dice+Focal loss adaptation; higher variance for the gated variants; a fixed-test (not k-fold) low-data protocol; and CBAM outperforming our Hybrid (the simpler channel attention is stronger). Per-slice significance overstates certainty due to correlated slices.

Future work. Extend to full 3D volumes (where the paper operated); test harder organs (e.g. pancreas) and thin/branching structures; run k-fold at each training size and more seeds to firm up the low-data comparison; and investigate why context-conditioning helps less than simple channel recalibration, to improve the Hybrid design.